BIOMEDICAL LIMITED, ILORIN, No 1 Ohimege Road, Industrial Estate, Ilorin Kwara State, PMB 1499.	RAW MATERIAL ANALYTICAL SPECIFICATIONS & METHODS	STP №	: BML/QA/STP/RMA/010/F02
		Material Code №	: RMA010
		Effective Date	: 4th Jan., 2024
		Supersedes	:
		Review Date	: 4th Dec., 2026
	POVIDONE	Revision No	: 01
		Ref. Book	BP 2023/IN-HOUSE
		Page	: 1 Of 14
Storage Conditions:	**Store in a well-closed container, protected from light**	**Department**	**: Quality**
Retest Frequency:	**12 months from date of release**	**Section**	**Quality Control**

ANALYTICAL METHODS:

1		**DESCRIPTION**
	1.1	Physically examine a representative sample of the material for Appearance, Aroma/Odour and flavour, and the presence of foreign matter: It is a White or yellowish-white powder or flakes, hygroscopic, freely soluble in water, in alcohol and in methanol, slightly soluble in acetone, practically insoluble in ether.
		NOTE:
	1.2	POVIDONE is poly[1-(2-oxo-1-pyrrolidinyl)ethylene] and consists of linear polymers of 1-vinylpyrrolidin-2-one. It contains not less than 11.5 per cent and not more than 12.8 per cent of nitrogen, calculated with reference to the anhydrous substance. The different types of povidone are characterised by their viscosity in solution, expressed as a K-value. The K-value of povidone having a stated K-value of 15 or less is not less than 85.0 per cent and not more than 115.0 per cent of the stated value. The K-value of povidone having a stated K-value or a stated K-value range with an average of more than 15 is not less than 90.0 per cent and not more than 108.0 per cent of the stated value or of the average of the stated range.
2		**Appearance Of Solution**
	2.1	Preparation Of Sample Solution1
	2.2	Add, through 100ml-measuring cylinder, 100ml of freshly distilled and cooled distilled water into a 250ml beaker
	2.3	Transfer onto magnetic stirrer, drop the bar-magnet and start stirring
	2.4	Accurately weigh and gradually add in small portions, while stirring, about 5g of sample.
	2.5	Observe the solution against bright northerly light: **Solution S is clear and not more intensely coloured than reference solution B6, BY6 or R6**
3		**Identification**
	3.1	<u>Preparation Of Sample Solution</u>
	3.1.1	Add, through 100ml-measuring cylinder, 100ml of freshly distilled and cooled distilled water into a 250ml beaker

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3.1.2 Transfer onto magnetic stirrer, drop the bar-magnet and start stirring

3.1.3 Accurately weigh and gradually add in small portions, while stirring , about 10g of sample.: **Sample dissolves completely**
Use this solution for all the identification test

3.2 Potassium Dichromate Test

3.2.1 Pipette 0.5ml of the sample solution into a test tube

3.2.2 Add 10ml of distilled water, 5ml of 2M hydrochloric acid, 2ml of 10.6%w/v potassium dichromate and mix: **An orange-yellow precipitate is formed.**

3.3 Dimethylaminobenzaldehyde Test

3.3.1 Pipette 1ml of this solution into a test tube

3.3.2 Add 0.2ml of dimethylaminobenzaldehyde solution and 0.1 ml of sulphuric acid and mix: **A pink colour is produced.**

3.4 Iodine Test

3.4.1 Pipette 0.1ml of this solution into a test tube

3.4.2 Add 5ml of distilled water, 0.2ml of 0.05M Iodine solution and mix: **A red colour is produced.**

4 pH

4.1 Procedure

4.2 Add, through 100ml-measuring cylinder, 100ml of freshly distilled and cooled distilled water into a 250ml beaker

4.3 Transfer onto magnetic stirrer, drop the bar-magnet and start stirring

4.4 Accurately weigh and gradually add in small portions, while stirring, about 5g of sample.

4.5 Measure the pH of the solution:

4.6 ***The pH of solution S is 3.0 to 5.0 for povidone having a stated K-value of at most 30.**

4.7 ***The pH of solution S is 4.0 to 7.0 for povidone having a stated K-value of more than 30.**

5 ALDEHYDES

5.1 500ppm Formaldehyde Standard Solution

5.1.1 Accurately weigh about 0.25g of analytical grade 40% formaldehyde stock into a 200ml volumetric flask.

5.1.2 Immediately before use, dilute with distilled water and mix. **Keep solution in the Refrigerator for not more than 3 months**

IF REQUIRED, DETERMINE THE ACTUAL CONCENTRATION AS FOLLOWS:

5.1.3 To 10 ml of the solution add 25 ml of 0.05M iodine VS and 10 ml of 1M sodium hydroxide.

5.1.4 Allow to stand for 5 minutes, add 11 ml of 1M hydrochloric acid and mix.

5.1.5 Titrate the excess iodine with 0.1M sodium thiosulphate VS using 1ml of 1%w/v starch mucilage added towards the end of the titration as indicator.

Carry out a blank titration omitting the sample.

Concentration (ppm)= (Blank Titre volume – Sample Titre volume) X Molarity of $Na_2S_2O_3$ X 1500

5.2 Sample Preparation Solution

5.2.1 Add, through 100ml-measuring cylinder, 100ml of freshly distilled and cooled distilled water into a 250ml beaker

5.2.2 Transfer onto magnetic stirrer, drop the bar-magnet and start stirring

5.2.3 Accurately weigh and gradually add in small portions, while stirring, about 5g of sample.

5.3 Decolourized Pararosaniline (basic fuchsin) Solution

5.3.1 Accurately weigh about 0.1 g of Pararosaniline into a 100ml volumetric flask.

5.3.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

5.3.3 Add 1g of anhydrous sodium sulphite dissolved in 10 ml of water and mix.

5.3.4 Slowly add 2ml of hydrochloric acid, stirring continuously.

5.3.5 Dilute to the mark with distilled water and mix.

5.3.6 Allow standing protected from light for at least 12 hours.

5.3.7 Add 0.3g of activated carbon and shake vigorously to remove the colour.

5.3.8 Filter immediately through Whatman # 1 filter paper rejecting the first 20ml of filtrate.

NOTE:

***If the solution becomes cloudy, filter before use.**

***If on standing the solution becomes violet, decolourize again by adding activated charcoal.**

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***Sensitivity Test:**
a] To 1.0 ml of the solution add 1.0 ml of water and 0.1 ml of aldehyde-free 96% ethanol.
b] Add 0.2 ml of a solution containing 0.01% w/v of formaldehyde: A pale pink colour is produced within 5 mins.

5.3 Procedure

5.3.1 Into three 25ml volumetric flasks, pipette and add in the following order:

Reagent s	Sample	Standard	Blank
Sample Solution	20.0ml	-	-
Reference Standard Solution	-	20.0ml	-
Water	-	-	20.0ml
Decolourized Pararosaniline Solution	2.0ml	2.0ml	2.0ml

5.3.2 Cover each flask and vigorously shake for 1 minute.

5.3.3 Allow standing at 24-26°C for 1 hour.

5.3.4 Using a suitable spectrophotometer and 1-cm cells, measure the absorbance of each solution at 552nm against the prepared blank: **The absorbance of the solution measured at 552 nm is not greater than that of a 500ppm reference standard**

6 Loss On Drying (Carry out in duplicate)

6.1 Weigh petri-dish (W1) that has previously been dried at 100°C and cooled.

6.2 Accurately weigh about 0.5-0.6g of sample onto each petri-dish (ensuring they are evenly distributed), then determine their weight together (W2).

6.3 Dry for 2 hours in an oven operating at 100°C.

6.4 Transfer into a suitable desiccator and allow cooling.

6.5 Reweigh (W3) the dried sample.

$$\% \text{ Loss On drying (\%LOD)} = \frac{(W2 - W3) \times 100}{(W2 - W1)}$$

Where,

W1 = weight of empty Petri-dish, g.
W2 = weight of sample and Petri-dish, g.
W3 = weight of Petri-dish and residue, g

7 SULPHATED ASH

7.1 Heat a silica or platinum crucible to redness for 30 minutes, allow cooling in a desiccator & weigh.

7.2 Weigh two crucibles (W1) that has previously been ignited and cooled.

7.3 Accurately weigh about 1g onto each crucible, ensuring they are evenly spread, then determine their weight together (W2).

7.4 Heat, gently at first on a Bunsen flame, until the sample is thoroughly charred.

7.5 Cool, and then moisten the residue with 1mL of sulphuric acid, heat gently until white fumes are no longer evolved, and ignite at 800 ± 25°C until the carbon is consumed (about 2 hours).

7.6 Cool in a desiccator, weigh, and calculate the % ROI, if the %ROI exceed 0.1%, again moisten the residue with 1mL of sulphuric acid, heat and ignite as before.

7.7 Continue until constant weight is attained (until two successive weighing does not differ by more than 0.5mg).
Weigh the crucibles with its content (W3).

Calculation:

$$\% \text{ Sulphated Ash} = \frac{(W3 - W1) \times 100}{W2 - W1}$$

Where, W1 = weight of empty crucible, g.
W2 = weight of sample and crucible, g.
W3 = weight of crucible and residue, g.

8 HEAVY METALS

8.1 Lead nitrate stock solution(100ppm)
Dissolve 160mg of lead nitrate, Analar, in 100ml distilled water to which has added 1ml concentrated nitric acid, Analar. Dilute with distilled water to 1000ml and mix. Prepare

and store this solution in glass containers free from soluble lead salts.

8.2 Lead nitrate stock solution (10ppm)

Pipette10.0ml of the 100ppm lead standard solution into a 100ml volumetric, dilute to the mark with distilled water and mix.

8.3 Buffer Solution pH3.5

In a 250ml beaker, dissolve 25.0g of ammonium acetate in 25ml of water and add 38.0ml of 7M hydrochloric acid.

Adjust the pH if necessary with 2M hydrochloric acid or 6M ammonium hydroxide and dilute to 100.0ml with water.

8.4 Thioacetamide SOLUTION(4%)

Dissolve 4g of thioacetamide in a 100ml volumetric flask, make up with distilled water Thioacetamide Reagent

To 2ml of thioacetamide solution in a 100ml volumetric flask, add 7.5ml of water, 37.5ml of M sodium hydroxide and 50ml of glycerol (85%). Heat in a water-bath for 20 seconds. Prepare immediately before use.

8.5 **Sample Preparation**

8.5.1 Accurately weigh and transfer about 2g of sample into a 20ml volumetric flask.

8.5.2 Add 15ml of distilled water and mechanically shake/sonicate till dissolution is complete.

8.5.3 Dilute to the mark with same solvent and mix.

8.5.4 To 12ml of the solution thus obtained (in a test tube), add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

8.6 Standard Preparation

8.6.1 Transfer 10 ml of 10ppm lead standard solution and 2 ml of the sample solution into a suitable test tube.

8.6.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

8.7 Blank Preparation

8.7.1 Prepare a blank using a mixture of 10ml of water and 2ml of sample solution.

8.7.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

OBSERVATION

Compared to the blank, after 2 minutes any brown colour in the solution obtained with the sample is not more intense than in the standard solution

9 PEROXIDES

9.1 13%v/v Sulphuric Acid

9.1.1 Measure 60ml of distilled water into a 100ml volumetric flask

9.1.2 Transfer into a cold water-bath.

9.1.3 Add, through the wall, 13ml of concentrated sulphuric acid, swirling from time to time make to the mark with distilled water

9.2 Titanium Trichloride-Sulphuric Acid Reagent

9.2.1 Into a 250ml beaker, transfer 20ml of 15%w/v titanium chloride solution.

9.2.2 Carefully , add through the wall of the beaker, 13ml of concentrated sulphuric acid and mix

9.2.3 Add sufficient hydrogen peroxide solution (100 vol) to produce a yellow colour

9.2.4 Transfer onto a hot plate and heat until white fumes are evolved.

9.2.5 Allow to cool. Dilute with water and repeat the evaporation and addition of water until a colourless solution is obtained.

9.2.6 Transfer into a 100ml volumetric flask; dilute this solution to 100 ml with water and mix.

9.3 Sample Preparation

9.3.1 Accurately weigh about 2g of sample into a 50ml volumetric flask.

9.3.2 Add 30ml of distilled water and mechanically shake/sonicate till dissolution is complete.

9.3.3 Dilute to the mark with distilled water and mix

9.4 Procedure

9.4.1 Into three 50ml volumetric flasks, pipette and add in the following order:

Reagent s	Sample	Blank
Sample Solution	25.0ml	25.0ml
13%v/v Sulphuric Acid	-	2.0ml
Titanium Trichloride-Sulphuric Acid Reagent	2.0ml	

9.4.2 Cover each flask and vigorously shake for 1 minute.

9.4.3 Using a suitable spectrophotometer and 1-cm cells, measure the absorbance of each solution at 405nm against the prepared blank: **The absorbance of the solution, measured at 405nm is not greater than 0.35 (400 ppm, expressed as H_2O_2).**

10 Vinylpyrrolidinone

10.1 Sample Preparation

10.1.1 Accurately weigh about 4g of sample each into separate 250ml conical flask.

10.1.2 Add 30ml of distilled water and mechanically shake/sonicate till dissolution is complete.

10.1.3 Add 0.5g of sodium acetate, swirl to dissolve and titrate with 0.1N iodine solution until the colour of iodine no longer fades. Then add additional 3.0ml extra volumes of the iodine solution.

10.1.4 Allow standing for 10 minutes

10.1.5 Titrate the excess iodine with 0.1n sodium thiosulphate to a faint yellow colour

10.1.5 Add 3ml of 1%w/v starch mucilage and continue titration to the disappearance of blue-black colour of the iodine-starch complex.

10.1.6 Carry out a blank titration omitting the sample

10.1.7 Volume of Iodine consumed by the sample

= (Blank Titre – Sample Titre) X Actual Molarity Of Sodium Thiosulphate X 10

Not more than 0.1ml of 0.05M Iodine is consumed, corresponding to not more than 0.001% (10ppm) vinylpyrrolidinone

11 Viscosity

11.1 Transfer about 80ml of distilled water into a 100ml standard flask.

11.2 Transfer onto magnetic stirrer, drop the bar-magnet and start stirring

11.3 Accurately weigh and gradually add in small portions, while stirring, about 10g of sample.

11.4 Mix till dissolution is complete and make up to 100ml.

11.5 Measure the viscosity after one hour ensuring that the temperature is between 24°C - 26°C and record the viscosity reading.

11.6 Estimate the relative viscosity of the distilled water and the product by comparison.
Relative Viscosity is 5.5 – 8.5.

12 Nitrogen

12.1 Method A

12.1.1 Screened Methyl Red Indicator
Accurately weigh about 75mg of bromocresol green and 50mg of methyl red into a 100ml volumetric flask
Add 60ml of ethanol, mechanically shake/sonicate till dissolution is complete
Dilute to the mark with distilled water and mix

12.1.2 0.05M sulphuric acid;prepared.

12.1.3 0.1M Sodium hydroxide ;prepared.

12.2 PROCEDURE

12.2.1 Accurately weigh about 50g of sample into a 100ml beaker

12.2.2 Transfer into a litre round-bottomed flask containing some glass beads or anti-bumping granules, using about 100ml of distilled water.

12.2.3 Transfer 100ml of 20% sodium or potassium hydroxide into the round-bottomed flask and cock immediately
Connect the distillation apparatus with the delivery end (immersed in the acid solution) into a conical flask containing 20.00ml of 0.05M sulphuric acid, 30ml of distilled water

12.2.4 Add 2 - 3 drops of screened Methyl Red Distil for 15 minute.

12.2.5 Rinse the condenser and the delivery end with distilled water, collecting the rinsing's in the conical flask.

12.2.6 Titrate the excess acid with 0.1M sodium hydroxide.

12.2.7 Simultaneously carry out a blank titration with 20.00ml of 0.05M sulphuric acid.

13 Microbiological Tests

13.1 <u>Limulus Amebocyte Lysate LAL Test</u>

13.1.1 Open the seals of the test tubes of lyophilized LAL Reagents

13.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

13.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

13.1.4 Incubate the reaction mixture immediately at 37°c for 60minutes. Avoiding vibration during incubation.

13.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

13.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

13.1.7 Read reaction tubes one by one.

13.1.8 Invert the tube in one smooth motion.

13.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by the absence of

solid clot after inversion.

Testing Methodology of Bacterial Endotoxin (LAL)

13.2.1 Each specimen shall be tested in duplicate

13.2.2 NMC: Take 100µl(0.1ml) LRW

13.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

13.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

13.2.5 Add the LAL reagents as mentioned below in the table 1

13.2.6 Mix gently and incubate the tubes in heating block at 37^0C ± 1 for 60min.

13.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 180^0C and note the observations.

13.2.8 If a firm gel has formed, record the results as positive.

13.2.9 A result is negative, if an intact gel is not formed.

13.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

13.3 **Pyrogen**

13.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed.

13.3.2 Withhold all food from the rabbits used during the period of the test

13.3.3 Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring.

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13.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit.

13.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm.

13.3.6 If the initial temperature is less than 38.8^{0}c or more than 39.8^{o}cexclude the animal from the test.

13.3.7 Ensure all the needles and syringe to be used are new

13.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 1^{0}c.

13.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection.

13.3.10 Any rabbits showing more than 0.2^{0}c temperature difference between two successive readings in the last three readings can be excluded from the test.

13.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes.

13.3.12 Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution.

13.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult

the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected.

13.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

13.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution

13.4 **MLT**

13.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

13.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

13.4.3 Testing to be performed under LAF of specified area

13.4.4 Arrange all the required material on the LAF before starting the test.

13.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

13.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45°C) in two petri plates for TAMC

13.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45°C) in two petri- plates for TYMC

13.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample,

allow medium to solidify, taking the precaution of not spilling the media from the plate

13.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45°c and allow for solidification.

13.4.1 Incubate the plates at 30°C to 35°C for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 20^{0}C to 25°C for 5 days in an inverted position.

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